

King Fahd University of Petroleum and Minerals**Department of Mathematics***Student effort not Official. Generated by Ice.***Math 208 Major Exam I Term 253**

Original exam date: 2 July 2026. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

- (a) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 \geq 0$
- (b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1^2 + x_2^2 = 1$
- (c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 - x_3 + 2x_4 = 1$
- (d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 - 2x_2 = 0$ and $x_3 + 5x_4 = 0$
- (e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_3 = 0$

2. If the function $g(x)$ with $g(0) = 1$ makes the differential equation $[g(x) - x \cos(xy)] dy = [e^{2x} + y \cos(xy)] dx$ an exact differential equation, then $g(x) =$

- (a) e^{2x}
- (b) $\sin x + 1$
- (c) 1
- (d) $\cos(2x)$
- (e) $\cos x$

3. The sum of all values of m for which $y = x^m$ is a solution of the differential equation $x^2y'' - 7xy' + 15y = 0$ is

- (a) 8
- (b) 0
- (c) -2
- (d) -8
- (e) 2

4. A thermometer with an initial reading of 10° is brought into a room kept at a constant 30° . If the thermometer reads 20° after 2 minutes, what is the total time required for the thermometer to reach 25° ?

- (a) 4 minutes
- (b) 6 minutes
- (c) 8 minutes
- (d) 5 minutes
- (e) 3 minutes

5. If the vectors $\mathbf{u} = (1, k, 2)$, $\mathbf{v} = (2, 2, 0)$ and $\mathbf{w} = (1, 2, -2)$ in $\mathbb{R}^3$ are linearly dependent, then $k =$

- (a) 1
- (b) -1
- (c) 0
- (d) 2
- (e) -2

6. The general solution of the differential equation $\frac{dy}{dx} + y \cot x = \sin x$, $0 < x < \pi$ is

- (a) $4y \sin x = x - 2 \sin(2x) + c$
- (b) $4y \sin x = 2x - \sin(2x) + c$
- (c) $2y \sin x = x - \sin(2x) + c$
- (d) $2y \sin x = 2x + \sin(2x) + c$
- (e) $3y \sin x = 4x - \sin(2x) + c$

7. The solution of the linear differential equation $\frac{dy}{dx} = \frac{y}{2y \ln y + y - x}$ is given by

- (a) $xy = cx + y^2 \ln y$
- (b) $2xy = c + y^2 \ln y$
- (c) $xy = c + x^2 \ln y$
- (d) $xy = c + y^2 \ln y$
- (e) $2xy = cx + y^2 \ln y$

8. The general solution of the differential equation $(x^2 + 3xy + y^2) dx - x^2 dy = 0$ is given by

(a) $\ln|x| + \left(1 + \frac{y}{x}\right)^{-1} = c$

(b) $\ln|x| + \frac{y}{x} + 1 = c$

(c) $y = 2x \ln|x| + cx$

(d) $\ln|x| + \left(1 + \frac{x}{y}\right)^{-1} = c$

(e) $y = x \ln|x| + cx$

9. A general solution of the exact differential equation $\left(\frac{2x}{y} - \frac{3y^2}{x^4}\right) dx + \left(\frac{2y}{x^3} - \frac{x^2}{y^2} + \frac{1}{\sqrt{y}}\right) dy = 0$ is

(a) $\frac{x^2}{y} - \frac{y^2}{x^2} + 2\sqrt{y} = c$

(b) $\frac{x}{y} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

(c) $\frac{x^2}{y} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

(d) $\frac{x^2}{y} - \frac{y}{x^3} + \sqrt{y} = c$

(e) $\frac{x^2}{y^2} + \frac{y^2}{x^3} + \sqrt{y} = c$

10. The general solution of the differential equation $xy'' + y' = 4x$, $x > 0$ is given by

(a) $y(x) = 2x^2 + Ax \ln x + B$

(b) $y(x) = 2x^2 + A \ln x + B$

(c) $y(x) = 4x^2 + Ax \ln x + B$

(d) $y(x) = x^2 + A \ln x + B$

(e) $y(x) = 3x^2 + A \ln x + B$

11. If a particle is moving in a straight line with acceleration $a(t) = \frac{1}{\sqrt{1-t^2}}$, initial position $x(0) = 1$, and initial velocity $v(0) = 0$, then the position of the particle at any time t is given by

(a) $x(t) = t \sin^{-1} t + 3\sqrt{1-t^2}$

(b) $x(t) = 2t \sin^{-1} t + 4\sqrt{1-t^2}$

(c) $x(t) = t \sin^{-1} t + \sqrt{1-t^2}$

(d) $x(t) = t \sin^{-1} t + 2\sqrt{1-t^2}$

(e) $x(t) = 2t \sin^{-1} t + \sqrt{1-t^2}$

12. If the solution space of the system

$$\begin{aligned}x_1 - 2x_2 + x_3 + x_4 &= 0 \\x_1 - 2x_2 + 2x_3 - x_4 &= 0\end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s, t are real numbers and $\mathbf{u} = (a, b, 0, 0)$, $\mathbf{v} = (m, n, 2, l)$, then $\frac{a}{b} + \frac{2}{l} =$

- (a) 2
- (b) 1
- (c) 0
- (d) 4
- (e) 3

13. By using an appropriate substitution, the differential equation $\frac{dy}{dx} = 1 + \sqrt{y - 4x + 1}$ transforms into the separable differential equation

- (a) $\frac{du}{\sqrt{u} + 3} = dx$
- (b) $\frac{du}{\sqrt{u} - 3} = dx$
- (c) $\frac{du}{\sqrt{u} + 2} = 3 dx$
- (d) $\frac{du}{\sqrt{u} + 1} = dx$
- (e) $\frac{du}{\sqrt{u} - 1} = 2 dx$

14. The solution of the initial-value problem $x \frac{dy}{dx} - y = y^2 \ln x$, $y(1) = \frac{1}{2}$ is given by

- (a) $\frac{1}{y} = 1 - 2 \ln x + \frac{1}{x}$
- (b) $\frac{1}{y} = 1 + 2 \ln x + \frac{1}{x}$
- (c) $\frac{1}{y} = 1 - \ln x + \frac{1}{x}$
- (d) $y = 1 + 2 \ln x - \frac{1}{2x}$
- (e) $y = 1 - \ln x + \frac{1}{x}$

15. If $y = y(x)$ is the solution of the initial value problem $\left(\frac{2 + \sin x}{y + 1}\right) \frac{dy}{dx} = -\cos x$, $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ equals

(a) $-\frac{1}{3}$

(b) $\frac{1}{3}$

(c) $-\frac{2}{3}$

(d) $\frac{4}{3}$

(e) $\frac{2}{3}$

Answer Key

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Q	Ans	Brief explanation
1	(d)	Homogeneous linear system; the others miss $\mathbf{0}$ or fail a closure.
2	(c)	$M_y = N_x$ forces $g'(x) = 0$; $g(0) = 1$.
3	(a)	$m^2 - 8m + 15 = 0$: $m = 3, 5$.
4	(a)	$T = 30 - 20e^{-kt}$, $e^{-2k} = \frac{1}{2}$, need $e^{-kt} = \frac{1}{4}$.
5	(c)	$\det = 4k = 0$.
6	(b)	$\rho = \sin x$: $(y \sin x)' = \sin^2 x = \frac{1 - \cos 2x}{2}$.
7	(d)	Linear in x : $(xy)'_y = 2y \ln y + y$, integral $y^2 \ln y$.
8	(a)	$v = y/x$: $xv' = (v + 1)^2$, so $-\frac{1}{v+1} = \ln x - c$.
9	(c)	$F = \frac{x^2}{y} + \frac{y^2}{x^3} + g(y)$, $g' = y^{-1/2}$.
10	(d)	$(xp)' = 4x$, $p = 2x + \frac{A}{x}$.
11	(c)	$v = \sin^{-1} t$; by parts $x = t \sin^{-1} t + \sqrt{1 - t^2} + C$, $C = 0$.
12	(d)	$\mathbf{u} \parallel (2, 1, 0, 0)$, and $x_3 = 2x_4$ gives $l = 1$.
13	(b)	$u = y - 4x + 1$: $u' = \sqrt{u} - 3$.
14	(c)	$v = 1/y$: $(xv)' = -\ln x$, $v = 1 - \ln x + \frac{C}{x}$, $C = 1$.
15	(b)	$y + 1 = \frac{K}{2 + \sin x}$, $K = 4$, so $y(\frac{\pi}{2}) = \frac{4}{3} - 1$.